

MATH 3220-002 Selected Solutions to PSet Problems

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In this document you may find solutions (proofs or examples) to selected problems from problem sets. As a general rule of thumb, it is more likely that this document will be updated with the solutions to problems that were graded on accuracy.

Throughout this class your solutions need not match the level of polish, precision and detail of the below solutions. What's most important is grasping the underlying concepts and steadily improving your problem-solving skills as well as your proof-writing and presentation skills.

Most solutions will have follow-up exercises. Just as in the problem sets, these follow-up exercises are stated in a neutral language, and determining whether or not the statement is correct is part of the exercise. A follow-up exercise here may later show up in an exam.

1 PSet 1, Problem 3

Let $f : V \rightarrow W$ be an arbitrary function between vector spaces.
Then f is linear iff its graph is a linear subspace of $V \times W$.

This statement is true without any modifications:

Claim: Let $f : V \rightarrow W$ be an arbitrary function between vector spaces. Then f is linear iff its graph is a linear subspace of $V \times W$.

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Proof: First note that the **Cartesian product** $V \times W$ of the vector spaces $V = (V, +_V, \cdot_V)$ and $W = (W, +_W, \cdot_W)$ itself is a vector space, with addition

$$+_{V \times W} : (V \times W) \times (V \times W) \rightarrow V \times W$$

and scalar multiplication

$$\cdot_{V \times W} : \mathbb{R} \times (V \times W) \rightarrow V \times W$$

defined entrywise, that is, for any $(v_1, w_1), (v_2, w_2) \in V \times W$, the vector sum is by definition

$$(v_1, w_1) +_{V \times W} (v_2, w_2) = (v_1 +_V v_2, w_1 +_W w_2),$$

and similarly, for any $(v, w) \in V \times W$ and for any scalar $c \in \mathbb{R}$, scalar multiplication is by definition

$$c \cdot_{V \times W} (v, w) = (c \cdot_V v, c \cdot_W w).$$

Thoroughout we'll use this linear structure on $V \times W$. As is customary, for brevity we'll suppress subscripts for vector addition and we'll not use any specific symbols for scalar multiplication.

($\Rightarrow$) First suppose $f : V \rightarrow W$ is linear. We claim that its graph is

a vector subspace of $V \times W$, that is, $\text{graph}(f)$ is closed under vector addition and scalar multiplication.

Let $(v_1, w_1), (v_2, w_2) \in \text{graph}(f)$, $c \in \mathbb{R}$. It suffices¹ to show that

$$c(v_1, w_1) + (v_2, w_2) = (cv_1 + v_2, cw_1 + w_2) \in \text{graph}(f).$$

By the definition of the graph of a function, this is true iff

$$cw_1 + w_2 = f(cv_1 + v_2).$$

Furthermore, we had taken the vectors $(v_1, w_1), (v_2, w_2)$ to be in the set $\text{graph}(f)$, so that we also have

$$w_1 = f(v_1), \quad w_2 = f(v_2).$$

Combining these equations with the previous one we obtain

$$cw_1 + w_2 = cf(v_1) + f(v_2) = f(cv_1 + v_2),$$

where in the last equality we used the linearity of f .

($\Leftarrow$) Now suppose $f : V \rightarrow W$ is an arbitrary function such that $\text{graph}(f)$ is a vector subspace of $V \times W$. We claim that f is in fact linear. For this it suffices² to show that for any $v_1, v_2 \in V$ and for any $c \in \mathbb{R}$, we have

$$f(cv_1 + v_2) = cf(v_1) + f(v_2).$$

By the definition of the graph of a function, we know that

$$(v_1, f(v_1)), (v_2, f(v_2)) \in \text{graph}(f),$$

and since we are assuming $\text{graph}(f)$ to be a linear subspace, it

¹See Follow-up Exercise 1 below.

²By Follow-up Exercise 1 below.

must be closed under linear combinations, hence

$$(cv_1 + v_2, cf(v_1) + f(v_2)) = c(v_1, f(v_1)) + (v_2, f(v_2)) \in \text{graph}(f).$$

Again by the definition of the graph of a function, it must be the case that

$$f(cv_1 + v_2) = cf(v_1) + f(v_2).$$

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This statement leads to a more general idea: what type of properties of a function can one recover by investigating its graph instead, and under which conditions? For instance the statement we established guarantees that the linearity of a function is equivalent to the linearity of its graph. Such statements in general are called "closed graph theorems"³.

³See Terence Tao's [blog post](#) surveying closed graph theorems in different contexts.

Follow-up Exercise 1

- (i) Let V and W be vector spaces, $f : V \rightarrow W$ be a function. Then f is linear iff

$$\forall v_1, v_2 \in V, \forall c \in \mathbb{R} : f(cv_1 + v_2) = cf(v_1) + f(v_2).$$

- (ii) Let V be a vector space, $S \subseteq V$ be a subset. Then S is a linear subspace iff

$$\forall v_1, v_2 \in S, \forall c \in \mathbb{R} : cv_1 + v_2 \in S.$$

Follow-up Exercise 2 Let V and W be vector spaces. Then

- (i) There are nonlinear functions $f : V \rightarrow W$ such that $\text{graph}(f)$ is closed under vector addition but not necessarily scalar multiplication.
- (ii) There are nonlinear functions $f : V \rightarrow W$ such that $\text{graph}(f)$ is closed under scalar multiplication but not necessarily vector addition.

Follow-Up Exercise 3 Let V and W be vector spaces. Then

- (i) Any proper linear subspace X of $V \times W$ is the graph of some function $f : V \rightarrow W$. Here **proper** means $X \neq V \times W$.
- (ii) Any proper linear subspace of V is the graph of some function $f : L \rightarrow R$ for some linear subspaces L, R of V such that **for any vector $v \in V$, there are unique $l \in L$ and $r \in R$ with $v = l + r$.**

2 PSet 2, Problem 3

Let V, W and X be normed spaces. Then

- (i) For any $L \in \mathcal{L}(V; W)$ and for any $K \in \mathcal{L}(W; X)$, $|K \circ L| \leq |K||L|$.
- (ii) If V, W , and X are finite dimensional, then $1 = \inf\{C \in \mathbb{R}_{\geq 0} \mid \forall L, K : |K \circ L| \leq C|K||L|\}$.

Let us examine the two parts separately. We'll use subscripts for different norms to disambiguate different magnitudes.

The first part is correct without any modifications:

Claim: Let V, W and X be normed spaces. Then for any $L \in \mathcal{L}(V; W)$ and for any $K \in \mathcal{L}(W; X)$, $|K \circ L|_{\text{op}} \leq |K|_{\text{op}}|L|_{\text{op}}$.

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Proof: Fix two linear transformations $L : V \rightarrow W$ and $K : W \rightarrow X$. Note that by problem 2.a in **PSET1**, $K \circ L : V \rightarrow X$ too is a linear operator. If either K or L is the zero transformation, then the LHS in the inequality is zero, hence there is nothing to prove. If neither K nor L is zero, and further either K or L is an unbounded linear operator, the RHS in the inequality is ∞ , and there is nothing to prove again.

Thus suppose both K and L are bounded linear operators different from the zero transformation. By a discussion in class, we know that for any $v \in V$ and for any $w \in W$,

$$|L(v)|_W \leq |L|_{\text{op}}|v|_V, \quad |K(w)|_X \leq |K|_{\text{op}}|w|_W.$$

Let $v \in V$. Then

$$|(K \circ L)(v)|_X = |K(L(v))|_X \leq |K|_{\text{op}}|L(v)|_W \leq |K|_{\text{op}}|L|_{\text{op}}|v|_V.$$

Thus by the definition of the operator norm, we have

$$|K \circ L|_{\text{op}} \leq |K|_{\text{op}} |L|_{\text{op}}.$$

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Next we discuss the second part. Let us first consider the Euclidean case, that is, the normed spaces involved are all spaces of tuples of real numbers, with norm induced by the standard dot product. In this case, say considering the standard bases, we can simply take the linear operators to be matrices of appropriate dimension. We'll also assume that all spaces involved are of at least dimension 1; indeed if any one of the spaces involved is zero dimensional (thus consisting of the zero vector only), then the statement fails (see below).

Claim: Let $V = \mathbb{R}^n$, $W = \mathbb{R}^m$ and $X = \mathbb{R}^p$ for n, m, p positive integers. Then

$$\inf \left\{ C \in \mathbb{R}_{\geq 0} \mid \forall L \in \mathcal{M}(m \times n; \mathbb{R}), \forall K \in \mathcal{M}(p \times m; \mathbb{R}): \begin{array}{l} |KL|_{\text{op}} \leq C |K|_{\text{op}} |L|_{\text{op}} \end{array} \right\} = 1,$$

where the operator norms are induced by norms defined by dot products.

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Proof: Denote by α the infimum in the claim. Thus the claim is that $\alpha = 1$. The first claim already established that $\alpha \leq 1$. Thus it suffices to show $\alpha \geq 1$.

Suppose $\alpha < 1$. We'll obtain a contradiction by constructing two transformations L and K such that

$$|K \circ L|_{\text{op}} = 1, \quad |K|_{\text{op}} = 1, \quad |L|_{\text{op}} = 1.$$

Define L to be the $m \times n$ matrix such that, if L_{ij} is the ij -th entry of

L , then $L_{ij} = 1$ if $i = 1 = j$ and $L_{ij} = 0$ otherwise. Similarly define K to be the $p \times m$ matrix with 1 in the top left entry and zero otherwise. Then a straightforward calculation shows that KL is the $p \times n$ matrix with 1 in the top left entry and zero in all other entries.

Let us now compute the operator norms of K , L and KL . Note that in fact **wlog** it suffices to compute the operator norm of one of these matrices. Let $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$. Then

$$|Lv| = |\underbrace{(v_1, 0, 0, \dots, 0)}_{\in \mathbb{R}^m}| = |v_1| \leq \left(\sum_{i=1}^n |v_i|^2 \right)^{1/2} = |v|.$$

Hence $|L|_{\text{op}} \leq 1$. Further, for $v = (1, 0, \dots, 0) \in \mathbb{R}^n$, we have $|Lv| = 1 = |v|$, so that $|L|_{\text{op}} = 1$. Similarly the operator norms of both matrices K and KL are 1. By the definition of α , we have

$$1 = |KL|_{\text{op}} \leq \alpha |K|_{\text{op}} |L|_{\text{op}} = \alpha \leq 1,$$

a contradiction. ┘

A word regarding the construction used in the previous proof is in order. The construction is based on the heuristic that the operator norm of a linear transformation is the maximal stretch it can cause. In the above proof, we considered L and K to be the transformations that "delete" all but the first entries. This means that, effectively these two transformations "stretch" vectors along the "first axes" by a factor of 1, and completely collapse all other directions.

Let us next consider the "trivial" case:

Claim: Let $V = \mathbb{R}^n$, $W = \mathbb{R}^m$ and $X = \mathbb{R}^p$ for n, m, p nonnegative integers. If any one of n, m , or p is zero, then

$$\inf \left\{ C \in \mathbb{R}_{\geq 0} \mid \forall L \in \mathcal{M}(m \times n; \mathbb{R}), \forall K \in \mathcal{M}(p \times m; \mathbb{R}): \begin{array}{l} |KL|_{\text{op}} \leq C |K|_{\text{op}} |L|_{\text{op}} \end{array} \right\} = 0,$$

where the operator norms are induced by norms defined by dot products.

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Proof: We have three cases to consider: $n = 0$, $m = 0$, or $p = 0$. In any one of these cases, $K \circ L$ is necessarily the zero transformation, and since the operator norm is a norm, $|K \circ L|_{\text{op}} = 0$ and hence the infimum is zero.

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Finally we discuss the general version of the second part of the statement. While this too is true (assuming that none normed spaces are zero dimensional), with what we have developed so far we can not prove it as is. Instead we'll prove a version of the second part of the statement with two additional hypotheses (which hypothesis we'll later see in the course is redundant). The idea is similar to the construction we used before, namely, we'll use two rank 1 transformations whose stretch effects compound maximally. Indeed, ultimately the two additional hypotheses are there so that the construction for matrices work in the abstract case:

Claim: Let V, W , and X be finite but positive dimensional normed spaces. If

- there are nonzero bounded linear functions $V \rightarrow \mathbb{R}$ and $W \rightarrow \mathbb{R}$, and
- W has the property that

$$\forall \kappa \in \mathcal{BL}(W; \mathbb{R}), \exists w_0 \in W : |w_0|_W = 1, |\kappa(w_0)| = |\kappa|_{\text{op}},$$

then

$$\inf \left\{ C \in \mathbb{R}_{\geq 0} \mid \forall L \in \mathcal{L}(V; W), \forall K \in \mathcal{L}(W; X) : \frac{|KL|_{\text{op}}}{|K|_{\text{op}}|L|_{\text{op}}} \leq C \right\} = 1.$$

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Proof: We already have by the first claim that 1 is an upper bound on the infimum involved. Let $x_0 \in X$ be a nonzero vector, $\kappa : W \rightarrow \mathbb{R}$ be nonzero bounded linear (note that we assume there is such a κ) and define $K : W \rightarrow X$ by

$$K : w \mapsto \kappa(w)x_0.$$

Then due to the linear nature of κ , K is linear. Further, a straightforward calculation shows that

$$|K|_{\text{op}} = |\kappa|_{\text{op}}|x_0|_X.$$

By hypothesis, there is a $w_0 \in W$ such that

$$|w_0|_W = 1, \quad |\kappa(w_0)| = |\kappa|_{\text{op}}.$$

Let $\lambda : V \rightarrow \mathbb{R}$ be nonzero bounded linear (again we assume such a λ exists) and define $L : V \rightarrow W$ by

$$L : v \mapsto \lambda(v)w_0.$$

L too is linear and again by a calculation we have

$$|L|_{\text{op}} = |\lambda|_{\text{op}}|w_0|_W = |\lambda|_{\text{op}}.$$

Consequently we have

$$|K|_{\text{op}}|L|_{\text{op}} = |\kappa|_{\text{op}}|\lambda|_{\text{op}}|x_0|_X.$$

On the other hand, for $v \in V$ arbitrary,

$$K \circ L(v) = K(\lambda(v)w_0) = \lambda(v)K(w_0) = \lambda(v)\kappa(w_0)x_0 = \lambda(v)|\kappa|_{\text{op}}x_0.$$

Consequently,

$$|K \circ L(v)|_X = |\lambda(v)| |\kappa|_{\text{op}} |x_0|_X.$$

Optimizing over all $v \in V$, we obtain

$$|K \circ L|_{\text{op}} = |\lambda|_{\text{op}} |\kappa|_{\text{op}} |x_0|_X = |K|_{\text{op}} |L|_{\text{op}},$$

establishing that the infimum involved indeed is equal to 1.

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In closing, let us mention why the two additional hypotheses are redundant: First, any linear operator defined on a finite dimensional normed space turns out to be a bounded linear operator. Second, for any linear operator defined on a finite dimensional normed space, it turns out that there is an input vector with unit norm whose image under the linear operator has norm exactly the operator norm of the operator.

Follow-Up Exercise 1 Let n be a positive integer and $\mathcal{M}$ be the vector space of $n \times n$ matrices with real entries. Then for any norm $|\bullet|$ on $\mathcal{M}$, the following holds:

$$\forall A, B \in \mathcal{M} : |AB| \leq |A||B|.$$

Follow-Up Exercise 2 Let n be a positive integer and $\mathcal{M}$ be the vector space of $n \times n$ matrices with real entries. Then for any norm $|\bullet|$ on $\mathcal{M}$, the following holds:

$$\exists c \in \mathbb{R}_{\geq 0}, \forall A, B \in \mathcal{M} : |AB| \leq c|A||B|.$$

Follow-Up Exercise 3 Let n be a positive integer and $\mathcal{M}$ be the vector space of $n \times n$ matrices with real entries. Then

$$\exists A, B \in \mathcal{M} : |AB|_{\text{op}} \leq |A|_{\text{op}}|B|_{\text{op}}.$$

Follow-Up Exercise 4 Let V, W and X be normed spaces. Then

$$1 = \inf\{C \in \mathbb{R}_{\geq 0} \mid \forall L, K : |K \circ L| \leq C|K||L|\}.$$

3 PSet 3, Problem 2

Let V be a normed space and $v_\bullet$ be a sequence in V converging to 0. Then for any normed space W and any bounded sequence of bounded linear operators $B_\bullet$ from V to W , $\lim_{n \rightarrow \infty} B_n(v_n) = 0$.

This statement is indeed true without any modifications:

Claim: Let V be a normed space and $v_\bullet$ be a sequence in V converging to 0. Then for any normed space W and any bounded sequence of bounded linear operators $B_\bullet$ from V to W , $\lim_{n \rightarrow \infty} B_n(v_n) = 0$.

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Proof: For any n , we have by a result we proved in class,

$$0 \leq \|B_n(v_n)\|_W \leq \|B_n\|_{\text{op}} \|v_n\|_V \leq \underbrace{\left(\sup_n \|B_n\|_{\text{op}} \right)}_{< \infty} \|v_n\|_V.$$

Note that we have that $\sup_n \|B_n\|_{\text{op}}$ is a finite number by the assumption that $B_\bullet$ is a *bounded sequence* of bounded operators. As $v_\bullet$ converges to the zero vector in V , by the continuity of norms, the sequence $\|v_\bullet\|_V$ of scalars converges to the zero number. Thus by the fact that a scalar multiple of a sequence of scalars that converges to the zero number converges to zero itself, and the squeeze theorem, we are done.

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Follow-Up Exercise Let V and W be normed spaces and $B_\bullet$ be a sequence of bounded linear operators from V to W converging to 0. Then for any bounded sequence $v_\bullet$ in V , $\lim_{n \rightarrow \infty} B_n(v_n) = 0$.

4 PSet 4, Problem 2

Let X be a metric space and $A, B \subseteq X$. Then

- (i) $\text{clos}(A \cup B) = \text{clos}(A) \cup \text{clos}(B)$.
- (ii) $\text{clos}(A \cap B) = \text{clos}(A) \cap \text{clos}(B)$.
- (iii) $\text{int}(A \cup B) = \text{int}(A) \cup \text{int}(B)$.
- (iv) $\text{int}(A \cap B) = \text{int}(A) \cap \text{int}(B)$.

The set operations of closure and interior have many different (but equivalent) definitions. Throughout the discussion below let us use the definitions given in class⁴. Thus if $A \subseteq X$ is a subset of a metric space, then the closure $\text{clos}(A)$ of A is by definition the set of all those points in X that can be approximated arbitrarily well by points in A , and the interior $\text{int}(A)$ of A is by definition the set of all those points in A whose sufficiently small perturbations are in A . Formally, we have:

$$\begin{aligned}\text{clos}(A) &= \{x \in X \mid \forall \varepsilon \in \mathbb{R}_{>0} : X[x] < \varepsilon \cap A \neq \emptyset\}, \\ \text{int}(A) &= \{x \in X \mid \exists \varepsilon \in \mathbb{R}_{>0} : X[x] < \varepsilon \subseteq A\}.\end{aligned}$$

Recall that here $X[x] < \varepsilon$ is the open ball in X centered at x or radius ε , that is,

$$X[x] < \varepsilon = [x] < \varepsilon = \{x' \in X \mid d(x', x) < \varepsilon\}.$$

With these definitions in mind, let's investigate the four statements given. The first statement is correct without any modification:

⁴While many parts of the first problem in this problem set too can be used to study the statements in this problem, we decided to present a self-contained account to exemplify the type of argumentation based on first principles.

Claim: Let X be a metric space. Then

$$\forall A, B \subseteq X : \text{clos}(A \cup B) = \text{clos}(A) \cup \text{clos}(B).$$

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Proof: Fix two subsets $A, B \subseteq X$. To show the equality of LHS and RHS it suffices to show that LHS is a subset of RHS and RHS is a subset of LHS.

($\supseteq$) Let $x \in \text{clos}(A) \cup \text{clos}(B)$. Then either x is in the closure of A , or x is in the closure of B (or both). Thus either for any $\varepsilon \in \mathbb{R}_{>0}$, $X[x] < \varepsilon \cap A$ is nonempty, or for any $\varepsilon \in \mathbb{R}_{>0}$, $X[x] < \varepsilon \cap B$ is nonempty. Then in either case for any $\varepsilon \in \mathbb{R}_{>0}$,

$$\emptyset \subsetneq (X[x] < \varepsilon \cap A) \cup (X[x] < \varepsilon \cap B) = X[x] < \varepsilon \cap (A \cup B),$$

as intersections distribute over unions. Thus x is in the closure of $A \cup B$.

($\subseteq$) We'll argue **contrapositively**. Let $x \notin \text{clos}(A) \cup \text{clos}(B)$. Then by **de Morgan's laws**, x is neither in the closure of A nor is it in the closure of B . Thus there are $\alpha, \beta \in \mathbb{R}_{>0}$ such that

$$X[x] < \alpha \cap A = \emptyset = X[x] < \beta \cap B.$$

Put $\varepsilon = \min\{\alpha, \beta\}$. Then again by the distributivity of intersections over unions

$$\begin{aligned} X[x] < \varepsilon \cap (A \cup B) &= (X[x] < \varepsilon \cap A) \cup (X[x] < \varepsilon \cap B) \\ &= \emptyset \cup \emptyset = \emptyset, \end{aligned}$$

so that x is not in the closure of $A \cup B$.

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The second statement is not quite correct; indeed in general it works only in one direction:

Claim: Let X be a metric space. Then

$$\forall A, B \subseteq X : \text{clos}(A \cap B) \subseteq \text{clos}(A) \cap \text{clos}(B).$$

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Proof: Fix two subsets $A, B \subseteq X$. Let x be in the closure of $A \cap B$. Thus by the monotonicity of intersections, for any $\varepsilon \in \mathbb{R}_{>0}$,

$$\emptyset \subsetneq X[x] < \varepsilon \cap (A \cap B) \subseteq X[x] < \varepsilon \cap A,$$

so that x is in the closure of A , as well as

$$\emptyset \subsetneq X[x] < \varepsilon \cap (A \cap B) \subseteq X[x] < \varepsilon \cap B,$$

so that x is in the closure of B also.

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Claim: There are subsets A, B in some metric space X such that

$$\text{clos}(A \cap B) \subsetneq \text{clos}(A) \cap \text{clos}(B).$$

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Proof: To prove this claim it suffices to exhibit an example. For instance consider $X = \mathbb{R}$, $A = [-1, 0[$, $B =]0, 1]$. Then $A \cap B$ is empty, hence so is its closure. On the other hand the closure of A is $[-1, 0]$ and the closure of B is $[0, 1]$, so that the intersection of the closures of A and B is precisely the **singleton** $\{0\}$.

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The next two statements provided are about the interior operation,

and we can study them either using the definitions or **duality**:

Lemma 1 (Duality): Let X be a metric space. Then

$$\forall A \subseteq X : \text{int}(A^c) = \text{clos}(A)^c.$$

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Here, for $B \subseteq X$, $B^c = X \setminus B$ is the **complement** of the set B inside X ; thus in **set builder notation**, $B^c = \{x \in X \mid x \notin B\}$. Note that the duality lemma extends the duality between unions and intersections established by complementation, that is, the de Morgan laws, to the interior and closure operations.

Proof (of Duality Lemma): Fix a subset $A \subseteq X$. By careful argumentation we can show the equality of two sets by chaining logical equivalences as follows:

$$\begin{aligned} x \in \text{int}(A^c) & \\ \Leftrightarrow \exists \varepsilon \in \mathbb{R}_{>0} : X[x] < \varepsilon \subseteq A^c & \\ \Leftrightarrow \exists \varepsilon \in \mathbb{R}_{>0} : X[x] < \varepsilon \cap A = \emptyset & \\ \Leftrightarrow x \notin \text{clos}(A) & \\ \Leftrightarrow x \in \text{clos}(A)^c. & \end{aligned}$$

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Claim: Let X be a metric space. Then

$$\forall A, B \subseteq X : \text{int}(A \cap B) = \text{int}(A) \cap \text{int}(B).$$

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We'll prove this claim in two ways: first using the duality lemma,

and second using the definition of interior.

Proof (using duality): Let $A, B \subseteq X$. Then we have

$$\begin{aligned}
 \text{int}(A \cap B) &\stackrel{*}{=} \text{int} \left((A \cap B)^c \right) \\
 &\stackrel{\dagger}{=} \text{clos} \left((A \cap B)^c \right)^c \\
 &\stackrel{\diamond}{=} \text{clos} \left(A^c \cup B^c \right)^c \\
 &= \left(\text{clos} \left(A^c \right) \cup \text{clos} \left(B^c \right) \right)^c \\
 &\stackrel{\diamond}{=} \text{clos} \left(A^c \right)^c \cap \text{clos} \left(B^c \right)^c \\
 &\stackrel{\dagger}{=} \text{int} \left(A^{cc} \right) \cap \text{int} \left(B^{cc} \right) \\
 &\stackrel{*}{=} \text{int}(A) \cap \text{int}(B),
 \end{aligned}$$

where in the equalities marked with $*$, $\dagger$, $\diamond$ we used the double complement producing the original set, duality lemma, and de Morgan's law, respectively, and in the unmarked equality we used the distributivity of closure over unions we established before.

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Proof (using definitions): Fix two subsets $A, B \subseteq X$.

($\subseteq$) Let x be in the interior of $A \cap B$. Then for some $\varepsilon \in \mathbb{R}_{>0}$, $X[x] < \varepsilon$ is a subset of $A \cap B$. Since $A \cap B$ is a subset of A , by the **transitivity** of the "subset of" relation, $X[x] < \varepsilon$ is a subset of A , so that x is in the interior of A , and similarly this same open ball is a subset of B , so that x is in the interior of B also. Consequently $x \in \text{int}(A) \cap \text{int}(B)$.

($\supseteq$) Let $x \in \text{int}(A) \cap \text{int}(B)$. Thus x is in the interior of both A and B . Thus there are $\alpha, \beta \in \mathbb{R}_{>0}$ such that

$$X[x] < \alpha] \subseteq A, \quad \text{and} \quad X[x] < \beta] \subseteq B.$$

Put $\varepsilon = \min\{\alpha, \beta\}$. Then

$$X[x] < \varepsilon] \subseteq X[x] < \alpha] \cap X[x] < \beta] \subseteq A \cap B,$$

so that x is in the interior of $A \cap B$.

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The remaining statement is not quite correct:

Claim: Let X be a metric space. Then

$$\forall A, B \subseteq X : \text{int}(A \cup B) \supseteq \text{int}(A) \cup \text{int}(B).$$

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While we could prove this claim using duality again, let's argue using the definitions only:

Proof: Fix $A, B \subseteq X$. Let $x \in \text{int}(A) \cup \text{int}(B)$. Then x is either in the interior of A , or it is in the interior of B (or both). In the first case, for some $\alpha \in \mathbb{R}_{>0}$, $X[x] < \alpha] \subseteq A \subseteq A \cup B$, and similarly in the second case for some $\beta \in \mathbb{R}_{>0}$, $X[x] < \beta] \subseteq B \subseteq A \cup B$, so that in either case we are able to find an open ball centered at x completely contained in $A \cup B$, establishing that x is in the interior of $A \cup B$.

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Claim: There are subsets A, B in some metric space X such that

$$\text{int}(A \cup B) \not\supseteq \text{int}(A) \cup \text{int}(B).$$

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Proof: It suffices to exhibit an example. By duality, the complements of the subsets in our previous example ought to work. Indeed, say

$X = \mathbb{R}$, $A^c = [-1, 0[$, $B^c =]0, 1]$, so that

$$A =]-\infty, -1[\cup [0, \infty[; \quad B =]-\infty, 0] \cup]1, \infty[.$$

Then $A \cup B = \mathbb{R}$, hence so is its interior. On the other hand the interior of A is $]-\infty, -1[\cup]0, \infty[$ and the interior of B is $]-\infty, 0[\cup]1, \infty[$, so that the union of the interiors of A and B is precisely $\mathbb{R} \setminus \{0\}$.

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Follow-up Exercise 1 There are metric spaces X such that

$$\forall A, B \subseteq X : \text{clos}(A \cap B) = \text{clos}(A) \cap \text{clos}(B).$$

Follow-up Exercise 2 Let X be a metric space. Then for any subset $S \subseteq X$, there are subsets $A, B \subseteq X$ such that

$$S = (\text{clos}(A) \cap \text{clos}(B)) \setminus \text{clos}(A \cap B)$$

Follow-up Exercise 3 Let A be a set and for any $\alpha \in A$, let S_α be a subset of a metric space X . Then

- (i) $\text{clos}(\bigcup_{\alpha \in A} S_\alpha) = \bigcup_{\alpha \in A} \text{clos}(S_\alpha).$
- (ii) $\text{clos}(\bigcap_{\alpha \in A} S_\alpha) = \bigcap_{\alpha \in A} \text{clos}(S_\alpha).$
- (iii) $\text{int}(\bigcup_{\alpha \in A} S_\alpha) = \bigcup_{\alpha \in A} \text{int}(S_\alpha).$
- (iv) $\text{int}(\bigcap_{\alpha \in A} S_\alpha) = \bigcap_{\alpha \in A} \text{int}(S_\alpha).$

Follow-Up Exercise 4 Let X be a metric space. Then

- (i) If two open balls in X intersect, then their intersection contains an open ball.
- (ii) For any $x, y \in X$, any $\alpha, \beta \in \mathbb{R}_{>0}$ and any $z \in [x] < \alpha] \cap [y] < \beta]$:

$$\begin{aligned} \sup\{\gamma \in \mathbb{R}_{>0} \mid [z] < \gamma] \subseteq [x] < \alpha] \cap [y] < \beta] \} \\ = \min\{\alpha - d(z, x), \beta - d(z, y)\}. \end{aligned}$$

- (iii) For any $x, y \in X$ and any $\alpha, \beta \in \mathbb{R}_{>0}$, consider the function

$$\gamma = \gamma_{x,y,\alpha,\beta} : X \rightarrow \mathbb{R}_{\geq 0}, z \mapsto \min\{\alpha - d(z, x), \beta - d(z, y)\}.$$

Then $[x] < \alpha$ and $[y] < \beta$ has a nonempty intersection iff $\gamma_{x,y,\alpha,\beta}$ is not identically zero.

5 PSet 5, Problem 3

Let $M = \{(x, y) \in \mathbb{R}^2 \mid |x| \leq 10, |y| \leq 10\}$ and consider

$f : M \rightarrow \mathbb{R}^3$ defined by

$$(x, y) \mapsto (xy - 2x + y + 1, x^2 + y^2 + x - 3y + 6, xy^2).$$

Perform the following tasks:

- (i) Compute the derivative of f .
- (ii) Compute the derivative of f at $x = (1, -1)$.
- (iii) Compute the affine function $A : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ tangent to f at $(1, -1)$.
- (iv) Compute the directional derivative of f at $x = (1, -1)$ in the direction of $v = (15, 2)$.
- (v) Compute the derivative of f' .
- (vi) Compute the derivative of $(f')'$.

Let us denote the components of f by f_1, f_2 and f_3 , so that

$$f_1 : (x, y) \mapsto xy - 2x + y + 1,$$

$$f_2 : (x, y) \mapsto x^2 + y^2 + x - 3y + 6,$$

$$f_3 : (x, y) \mapsto xy^2.$$

M is open Recall that for differentiability we require (at least for now) that the function at hand is defined on an open subset of a normed space. First let us verify that indeed M is an open subset of $\mathbb{R}^2$. Let $(x, y) \in M$, so that $-10 \leq x \leq 10$ and $-10 \leq y \leq 10$. We need to show that for some $\delta \in \mathbb{R}_{>0}$, the open δ -ball centered at (x, y) is in M . Put

$$\delta = \min\{10 - |x|, 10 - |y|\} = 1 - \max\{|x|, |y|\} > 0,$$

and suppose $(\bar{x}, \bar{y}) \in \mathbb{R}^2$ is in the open δ -ball centered at (x, y) .
Then

$$\begin{aligned} |\bar{x}| - |x| &\leq ||\bar{x}| - |x|| \\ &\leq |\bar{x} - x| \\ &= \sqrt{(\bar{x} - x)^2} \\ &\leq \sqrt{(\bar{x} - x)^2 + (\bar{y} - y)^2} \\ &\leq \min\{10 - |x|, 10 - |y|\} \\ &\leq 10 - |x|, \end{aligned}$$

where in the second inequality we used the reverse triangular inequality. Cancelling $|x|$ from both sides we obtain $|\bar{x}| \leq 10$. Similarly we obtain $|\bar{y}| \leq 10$ also, so that $(\bar{x}, \bar{y}) \in M$.

Partial Derivatives Let us first record all partial derivatives of all components as we are familiar with from computation-based calculus classes. With higher order partial derivatives, it is likely that the order of differentiation will not matter, and we'll systematize the computation of partials like so: For each component f_j , we'll start a new chart. We'll put f_j to the bottom left corner. Then right next to it we'll put $\frac{\partial f_j}{\partial x}$ and right above it we'll put $\frac{\partial f_j}{\partial y}$. This way we'll populate the chart, one step to the right representing taking the x -partial and one step up representing taking the y -partial. Thinking of $\mathbb{Z}_{\geq 0}^2$ as a subset of the plane $\mathbb{R}^2$, this means that at the point $\alpha = (\alpha_1, \alpha_2)$ we have the $(\alpha_1 + \alpha_2)$ -th order partial derivative $\frac{\partial^{\alpha_1 + \alpha_2} f_j}{\partial x^{\alpha_1} \partial y^{\alpha_2}}$.

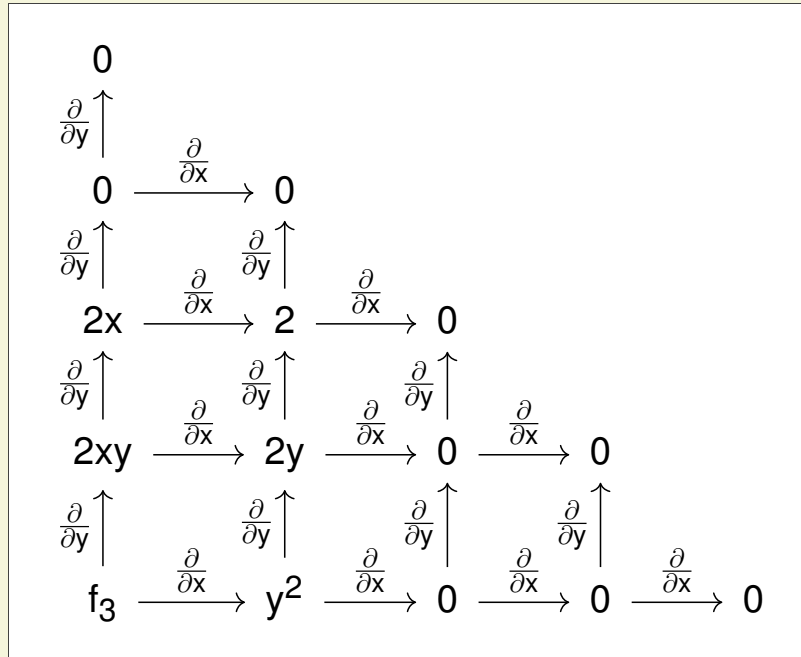
Thus the chart of partials of f_1 is:

$$\begin{array}{ccccccc}
 & & 0 & & & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial x} \rightarrow & & & \\
 & 0 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & \frac{\partial}{\partial x} \rightarrow & & \\
 x+1 & \xrightarrow{\frac{\partial}{\partial x}} & 1 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & \frac{\partial}{\partial y} \uparrow & \frac{\partial}{\partial x} \rightarrow & \\
 f_1 & \xrightarrow{\frac{\partial}{\partial x}} & y-2 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & \xrightarrow{\frac{\partial}{\partial x}} & 0
 \end{array}$$

Similarly the chart of partials of f_2 is:

$$\begin{array}{ccccccc}
 & & 0 & & & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial x} \rightarrow & & & \\
 & 2 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & \frac{\partial}{\partial x} \rightarrow & & \\
 2y-3 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & \frac{\partial}{\partial y} \uparrow & \frac{\partial}{\partial x} \rightarrow & \\
 f_2 & \xrightarrow{\frac{\partial}{\partial x}} & 2x+1 & \xrightarrow{\frac{\partial}{\partial x}} & 2 & \xrightarrow{\frac{\partial}{\partial x}} & 0
 \end{array}$$

Finally the chart of partials of f_3 is:



Algebraic Expansion Let us note that the function f is "of algebraic origin". That is, it is made up of three components and each component is a polynomial function in two variables⁵. Thus we can expect that simply plugging in sums and expanding out will be compatible with differential information. Let $(x, y) \in M$, $(h, k) \in \mathbb{R}^2$. We think of (x, y) as a position and (h, k) as a velocity or small perturbation. We'll plug in f the sum $(x + h, y + k)$, expand out the expressions and then organize the terms according to their orders in (h, k) :

⁵In fact, in the language we introduced the week after this problem set was due, we can say that f is a polynomial function.

$$\begin{aligned}
& f(x+h, y+k) \\
&= (f_1(x+h, y+k), f_2(x+h, y+k), f_3(x+h, y+k)) \\
&= ((x+h)(y+k) - 2(x+h) + (y+k) + 1, \\
&\quad (x+h)^2 + (y+k)^2 + (x+h) - 3(y+k) + 6, \\
&\quad (x+h)(y+k)^2) \\
&= (xy + xk + hy + hk - 2x - 2h + y + k + 1, \\
&\quad x^2 + 2xh + h^2 + y^2 + 2yk + k^2 + x + h - 3y - 3k + 6, \\
&\quad xy^2 + 2xyk + xk^2 + y^2h + 2yhk + hk^2) \\
&= \underbrace{(xy - 2x + y + 1, x^2 + y^2 + x - 3y + 6, xy^2)}_{\text{0th order}} \\
&\quad + \underbrace{((y-2)h + (x+1)k, (2x+1)h + (2y-3)k, y^2h + 2xyk)}_{\text{1st order}} \\
&\quad + \underbrace{(hk, h^2 + k^2, 2yhk + xk^2)}_{\text{2nd order}} \\
&\quad + \underbrace{(0, 0, hk^2)}_{\text{3rd order}}.
\end{aligned}$$

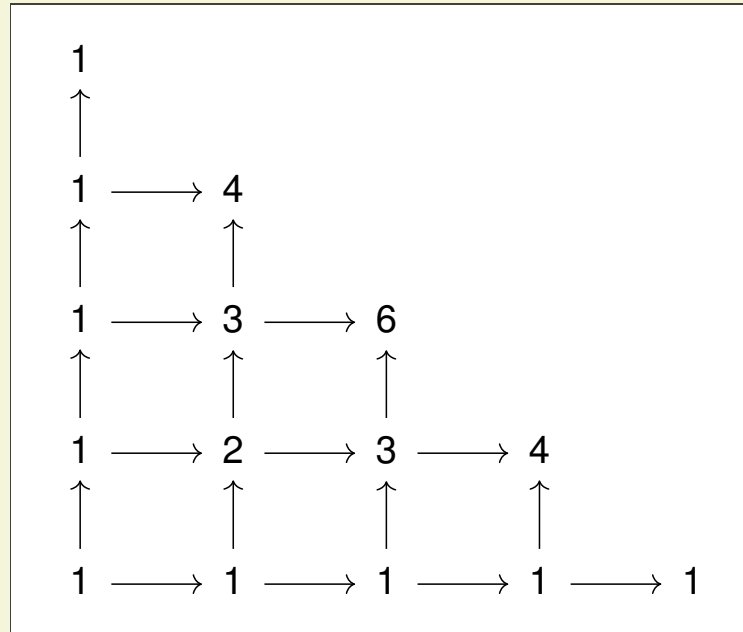
Here note that the 0th order term is exactly $f(x, y)$, that is, the 0th derivative of f at (x, y) . Similarly, the 1st order term ought to be related to the first derivative $f'(x, y)$ of f at (x, y) , the 2nd order term related to $f''(x, y)$ and 3rd order term to $f'''(x, y)$. The fact that there are no higher order terms in (h, k) ought to be related to the idea that all higher order derivatives of f are all zero.

Let us also observe the relation between the charts of partials we wrote above and the algebraic expansion. First focus on the tuple in the algebraic expansion that is of 1st order in (h, k) . In this tuple, the coefficients of h correspond exactly to the x -partials of

the components of f in the charts, and similarly the coefficients of k correspond exactly to the y -partials in the charts. In short, the variable $h = h^1 k^0$ corresponds to the $(1, 0)$ entries in the charts and the variable $k = h^0 k^1$ corresponds to the $(0, 1)$ entries in the charts. Note that this pattern applies to the 0th order term too: the variable $1 = h^0 k^0$ corresponds to the $(0, 0)$ entries in the charts.

Does the same pattern apply to higher order tuples in the algebraic expansion? Ultimately yes, but we'll need to reinterpret what the pattern actually is. Indeed, for instance focus on the tuple in the algebraic expansion that is of 2nd order in (h, k) . In this tuple, any component is made up of a linear combination of $h^2 = h^2 k^0$, $hk = h^1 k^1$ and $k^2 = h^0 k^2$. Again, it seems like the coefficients of $h^{\alpha_1} k^{\alpha_2}$ in the tuple correspond to the (α_1, α_2) entries in the charts, but at least for some coefficients there are multiplicative integer errors. I will leave it to you to how to reinterpret the pattern; for now it suffices to give two hints as to the cause of the multiplicative integer errors:

- (i) Much like in the single variable Taylor approximation, there should be correction factors: the k -th derivative ought to be corrected by $\frac{1}{k!}$ to match with the algebraic expansion.
- (ii) As the order of mixed partials ought not to matter, there should be correction factors due to different orders in which one can differentiate to match the algebraic expansion to the charts of partials; these correction factors should match with the **Pascal triangle** like so:



First Derivative To compute the first derivative of f' , we'll take the tuple in the algebraic expansion that is of first order in (h, k) , and write it as a linear function of (h, k) , for (x, y) fixed. Recall that our convention is that a linear transformation is encoded by a left multiplication by a rectangular matrix, so we'll write our tuples as column matrices. Thus

$$\begin{aligned}
 & \underbrace{((y-2)h + (x+1)k, (2x+1)h + (2y-3)k, y^2h + 2xyk)}_{\text{1st order}} \\
 &= \begin{pmatrix} (y-2)h + (x+1)k \\ (2x+1)h + (2y-3)k \\ y^2h + 2xyk \end{pmatrix} \\
 &= \underbrace{\begin{pmatrix} y-2 & x+1 \\ 2x+1 & 2y-3 \\ y^2 & 2xy \end{pmatrix}}_{=f'(x,y)} \begin{pmatrix} h \\ k \end{pmatrix} = f'(x, y)(h, k).
 \end{aligned}$$

Comparing with the charts of partials, we note that $f'(x, y)$ is the ma-

trix of first partials: row j corresponds to the component f_j , first column corresponds to the x -partials and the second column corresponds to the y -partials:

$$f'(x, y) = \begin{pmatrix} y - 2 & x + 1 \\ 2x + 1 & 2y - 3 \\ y^2 & 2xy \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x}(x, y) & \frac{\partial f_1}{\partial y}(x, y) \\ \frac{\partial f_2}{\partial x}(x, y) & \frac{\partial f_2}{\partial y}(x, y) \\ \frac{\partial f_3}{\partial x}(x, y) & \frac{\partial f_3}{\partial y}(x, y) \end{pmatrix}.$$

Finally let us do a **dimensional analysis**: according to our definition of the derivative, for each fixed $(x, y) \in M$, $f'(x, y)$ should be a linear transformation⁶ from $\mathbb{R}^2$ to $\mathbb{R}^3$, and hence the corresponding matrix should indeed be 3×2 .

Derivative at $(1, -1)$ Plugging in $(x, y) = (1, -1)$ in the 3×2 matrix corresponding to the derivative of f , we obtain

$$f'(1, -1) = \begin{pmatrix} -3 & 2 \\ 3 & -5 \\ 1 & -2 \end{pmatrix}.$$

Affine Approximation at $(1, -1)$ To compute the affine approximation of f at $(1, -1)$ we'll simply take the algebraic expansion, concatenate it by discarding the tuples corresponding to terms of order 2 or more, and then plugging in $(x, y) = (1, -1)$. Ultimately we'll obtain an affine function of (h, k) . In other words, from the point of view of calculus, the affine approximation⁷ at $(1, -1)$ is simply the function

⁶Recall that we proved in class that any linear transformation defined on a finite dimensional normed space is automatically bounded.

⁷Or, from the perspective of Taylor approximation, the **order 1 Taylor polynomial of f at $(1, -1)$**

$$T_1(f; (1, -1)) : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad (h, k) \mapsto f(1, -1) + f'(1, -1)(h, k).$$

Plugging $(x, y) = (1, -1)$ in $f(x, y)$, we obtain

$$\begin{aligned} f(1, -1) &= \underbrace{(xy - 2x + y + 1, x^2 + y^2 + x - 3y + 6, xy^2)}_{\text{0th order}}|_{(x,y)=(1,-1)} \\ &= (-1 - 2 - 1 + 1, 1 + 1 + 1 + 3 + 6, 1) \\ &= (-3, 12, 1). \end{aligned}$$

Thus in matrix notation the affine approximation of f at $(1, -1)$ is

$$T_1(f; (1, -1)) : \begin{pmatrix} h \\ k \end{pmatrix} \mapsto \begin{pmatrix} -3 \\ 12 \\ 1 \end{pmatrix} + \begin{pmatrix} -3 & 2 \\ 3 & -5 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix},$$

and in tuple notation we could write

$$T_1(f; (1, -1)) : (h, k) \mapsto (-3h + 2k - 3, 3h - 5k + 12, h - 2k + 1).$$

Directional Derivative at $(1, -1)$ in the Direction of $(15, 2)$ Since we know that f is differentiable everywhere, to compute its directional derivative at $(1, -1)$ in the direction of $(15, 2)$, we can simply compute $f'(1, -1)(15, 2)$:

$$f'(1, -1)(15, 2) = \begin{pmatrix} -3 & 2 \\ 3 & -5 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 15 \\ 2 \end{pmatrix} = \begin{pmatrix} -41 \\ 35 \\ 11 \end{pmatrix} = (-41, 35, 11).$$

We could have alternatively computed this directional derivative by taking a certain limit; I will leave the explicit calculation of the limit to you:

$$D_{(15,2)}f(1, -1) = \lim_{t \rightarrow 0} \frac{f(1 + 15t, -1 + 2t) - f(1, -1)}{t} = (-41, 35, 11).$$

Second Derivative Now we'll compute the second derivative of f , or more accurately the derivative of f' . Note that now we consider the function

$$f' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3), \quad (x, y) \mapsto \begin{pmatrix} y - 2 & x + 1 \\ 2x + 1 & 2y - 3 \\ y^2 & 2xy \end{pmatrix},$$

hence its derivative should be of the form

$$(f')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)).$$

Although f' takes values in a more complicated space compared to f , still it is of an "algebraic origin", so an algebraic expansion should suffice. Instead of using (h, k) we'll use $(\bar{h}, \bar{k})$ to denote the small perturbation:

$$\begin{aligned}
 f'(x + \bar{h}, y + \bar{k}) &= \begin{pmatrix} (y + \bar{k}) - 2 & x + \bar{h} + 1 \\ 2(x + \bar{h}) + 1 & 2(y + \bar{k}) - 3 \\ (y + \bar{k})^2 & 2(x + \bar{h})(y + \bar{k}) \end{pmatrix} \\
 &= \underbrace{\begin{pmatrix} y - 2 & x + 1 \\ 2x + 1 & 2y - 3 \\ y^2 & 2xy \end{pmatrix}}_{\text{0th order}} \\
 &\quad + \underbrace{\begin{pmatrix} \bar{k} & \bar{h} \\ 2\bar{h} & 2\bar{k} \\ 2y\bar{k} & 2x\bar{k} + 2y\bar{h} \end{pmatrix}}_{\text{1st order}} \\
 &\quad + \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ \bar{k}^2 & 2\bar{h}\bar{k} \end{pmatrix}}_{\text{2nd order}}.
 \end{aligned}$$

Here we wrote the orders *from the perspective of* f' . As the algebraic expansion of f has nonzero terms up to third order, in a compatible way the algebraic expansion of f' has nonzero terms up to second order.

For fixed $(x, y) \in M$, the function that takes the tuple $(\bar{h}, \bar{k})$ to the 3×2 matrix that is the first order component in the algebraic expansion is a linear transformation, and hence is the derivative $(f')'(x, y)$:

$$(f')'(x, y) : \mathbb{R}^2 \rightarrow \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3), \quad (\bar{h}, \bar{k}) \mapsto \begin{pmatrix} \bar{k} & \bar{h} \\ 2\bar{h} & 2\bar{k} \\ 2y\bar{k} & 2x\bar{k} + 2y\bar{h} \end{pmatrix}.$$

Obtaining this expression is a perfectly valid computation of the second derivative $(f')'(x, y)$ of f at (x, y) .

We can not quite write a matrix that corresponds to this linear transformation like we did for $f'(x, y)$, at least without **flattening**. Still, we can keep using matrix algebra by being a little flexible as to what its rules are. Indeed, the linearity of the 3×2 matrix $(f')'(x, y)(\bar{h}, \bar{k})$ in $(\bar{h}, \bar{k})$ is equivalent to the linearity of each of its 6 entries, and hence we can consider $(f')'(x, y)$ to be a 3×2 matrix each of whose entries is a 1×2 matrix: for $(\bar{h}, \bar{k}) \in \mathbb{R}^2$, we have

$$\begin{aligned}
 (f')'(x, y)(\bar{h}, \bar{k}) &= \begin{pmatrix} \bar{k} & \bar{h} \\ 2\bar{h} & 2\bar{k} \\ 2y\bar{k} & 2x\bar{k} + 2y\bar{h} \end{pmatrix} \\
 &= \begin{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} & \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\ \begin{pmatrix} 2 & 0 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\ \begin{pmatrix} 0 & 2y \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} & \begin{pmatrix} 2y & 2x \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \end{pmatrix} \\
 &= \begin{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 2 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} \\ \begin{pmatrix} 0 & 2y \end{pmatrix} & \begin{pmatrix} 2y & 2x \end{pmatrix} \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\
 &= \underbrace{\begin{pmatrix} 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \\ 0 & 2y & 2y & 2x \end{pmatrix}}_{=(f')'(x, y)} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix}.
 \end{aligned}$$

Note that in the last expression above, it seems like we are multiplying a 3×4 matrix with a 2×1 matrix, and we want this product to

be a 3×2 matrix. Certainly this is not in accordance with the standard matrix algebra conventions! Still, insofar as we understand and keep in mind what this "matrix product" actually stands for, we can safely use it⁸.

Again note that the entries of $(f')'(x, y)$ correspond to the second order partials in the charts. and again it is no coincidence that some of the entries are repeated. I will leave it to you to establish the pattern, only giving two hints:

- (i) Row j corresponds to the component f_j ,
- (ii) Second and third columns are exactly the same.

Second Derivative via Multilinear Algebra Here is another way to think about $(f')'(x, y)$: For fixed (x, y) , $(\bar{h}, \bar{k}) \mapsto (f')'(x, y)(\bar{h}, \bar{k})$ is a linear function of linear functions from $\mathbb{R}^2$ to $\mathbb{R}^3$. That is to say, we have a bilinear function $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^3$. So as to not confuse it with $(f')'(x, y)$, let us denote this bilinear function by $f''(x, y)$ ⁹:

$$f''(x, y) : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^3.$$

Thus we can feed two vectors (h, k) and $(\bar{h}, \bar{k})$ to it, and try to rearrange the resulting vector in $\mathbb{R}^3$. We'll use the key idea that to represent a scalar bilinear function we can let the matrix representation of $f''(x, y)$ to act from the left on one vector and from the right on the other vector:

⁸For instance addition and scalar multiplication work as usual, although not matrix multiplication. The formalism for these more complicated matrix products really is part of multilinear algebra, and it goes by names like "tensor product", "Kronecker product" or "outer product".

⁹As we discussed in class after the deadline for this problem set, this means that under the Currying isomorphism $\mathcal{L}_2(\mathbb{R}^2; \mathbb{R}^3) \cong \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3))$, $f''(x, y)$ transforms to $(f')'(x, y)$.

$$\begin{aligned}
f''(x, y)(\bar{h}, \bar{k})(h, k) &= \begin{pmatrix} \bar{k} & \bar{h} \\ 2\bar{h} & 2\bar{k} \\ 2y\bar{k} & 2x\bar{k} + 2y\bar{h} \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix} \\
&= \begin{pmatrix} \bar{k}h + \bar{h}k \\ 2\bar{h}h + 2\bar{k}k \\ 2y\bar{k}h + 2x\bar{k}k + 2y\bar{h}k \end{pmatrix} \\
&= \begin{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix}^T \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\ \begin{pmatrix} h \\ k \end{pmatrix}^T \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\ \begin{pmatrix} h \\ k \end{pmatrix}^T \begin{pmatrix} 0 & 2y \\ 2y & 2x \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \end{pmatrix} \\
&= \begin{pmatrix} h \\ k \end{pmatrix}^T \begin{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \\ \begin{pmatrix} 0 & 2y \\ 2y & 2x \end{pmatrix} \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\
&= \begin{pmatrix} h \\ k \end{pmatrix}^T \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 2 & 0 \\ 0 & 2 \\ 0 & 2y \\ 2y & 2x \end{pmatrix}}_{=f''(x,y)} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix}.
\end{aligned}$$

Again to make sense of the last equality we are forced to be more

flexible as to the rules of matrix algebra. Another important point is that the 2×2 entries of the 3×1 matrix representing $f''(x, y)$ are all **symmetric**; this indeed corresponds again to the fact that the order of mixed partials ought not matter.

Third Derivative The calculation of the third derivative of f , or more precisely the derivative of $(f')'$, is similar: Due to the algebraic nature of functions involved, an algebraic expansion should suffice. We start with the function

$$(f')' : M \rightarrow \mathcal{M}(3 \times 2; \mathcal{M}(1 \times 2; \mathbb{R})) \cong \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)),$$

$$(x, y) \mapsto \begin{pmatrix} 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \\ 0 & 2y & 2y & 2x \end{pmatrix}.$$

Let us use $(\tilde{h}, \tilde{k}) \in \mathbb{R}^2$ for the perturbation for the algebraic expansion:

$$\begin{aligned}
 (f')'(x + \tilde{h}, y + \tilde{k}) &= \begin{pmatrix} 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \\ 0 & 2(y + \tilde{k}) & 2(y + \tilde{k}) & 2(x + \tilde{h}) \end{pmatrix} \\
 &= \underbrace{\begin{pmatrix} 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \\ 0 & 2y & 2y & 2x \end{pmatrix}}_{\text{0th order}} \\
 &\quad + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2\tilde{k} & 2\tilde{k} & 2\tilde{h} \end{pmatrix}}_{\text{1st order}}.
 \end{aligned}$$

As before, the orders of the terms is from the perspective of $(f')'$; hence the k -th order term in $(f')'$ corresponds to the $(k + 1)$ -st order term in the expansion of f' , which in turn corresponds to the $(k + 2)$ -nd order term in the expansion of f .

We again note that the first order term in the algebraic expansion is a linear function of $(\tilde{h}, \tilde{k})$, and that linear function is $((f')')'(x, y)$. Relaxing the rules of matrix algebra, we can again obtain a matrix representation for the third derivative, this time obtaining a 3×2 matrix each of whose entries is a 1×2 matrix, with 1×2 matrix entries:

$$\begin{aligned}
((f')')'(x, y)(\tilde{h}, \tilde{k}) &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2\tilde{k} & 2\tilde{k} & 2\tilde{h} \end{pmatrix} \\
&= \begin{pmatrix} \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 2 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} \end{pmatrix} \\
&= \begin{pmatrix} \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} & \begin{pmatrix} 2 & 0 \end{pmatrix} \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} \\
&= \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 2 & 2 & 0 \end{pmatrix}}_{=((f')')'(x, y)} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix}.
\end{aligned}$$

Again I am leaving it to you to determine the patterns.

Third Derivative via Multilinear Algebra We could compute the second derivative two ways:

- (i) $(f')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; (\mathcal{L}(\mathbb{R}^2; \mathbb{R}^3))) \cong \mathcal{M}(3 \times 2; \mathcal{M}(1 \times 2; \mathbb{R}))$, and
- (ii) $f'' : M \rightarrow \mathcal{L}_2(\mathbb{R}^2; \mathbb{R}^3) \cong \mathcal{M}(3 \times 1; \mathcal{M}(2 \times 2; \mathbb{R}))$.

Similarly there are multiple ways to compute the third derivative:

- (i) $((f')')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)))$,

$$(ii) \ (f')'' : M \rightarrow \mathcal{L}_2(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)),$$

$$(iii) \ (f'')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathcal{L}_2(\mathbb{R}^2; \mathbb{R}^3)),$$

$$(iv) \ f''' : M \rightarrow \mathcal{L}_3(\mathbb{R}^2; \mathbb{R}^3).$$

Above we really computed $(f')''$, by way of the isomorphism

$$\mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3))) \cong \mathcal{M}(3 \times 2; \mathcal{M}(1 \times 2; \mathcal{M}(1 \times 2; \mathbb{R}))).$$

Ultimately all of these functions ought to be compatible; I am leaving it to you to actually compute these different looking third derivatives; for instance for the third option you would differentiate the following function:

$$f'' : M \rightarrow \mathcal{M}(3 \times 1; \mathcal{M}(2 \times 2; \mathbb{R})) \cong \mathcal{L}_2(\mathbb{R}^2; \mathbb{R}^3),$$

$$(x, y) \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 2 & 0 \\ 0 & 2 \\ 0 & 2y \\ 2y & 2x \end{pmatrix}.$$

Fair warning: flexing the rules of matrix algebra to encode trilinearity for f''' will likely prove to be harder than the other computations.

Follow-up Exercise Find and correct all typos and computation errors in the solution above.

Follow-up Exercise 2 Complete all calculations left to you in the solution above.

Follow-up Exercise 3 Redo all calculations in the solution above by flattening the objects involved into column matrices at each step.